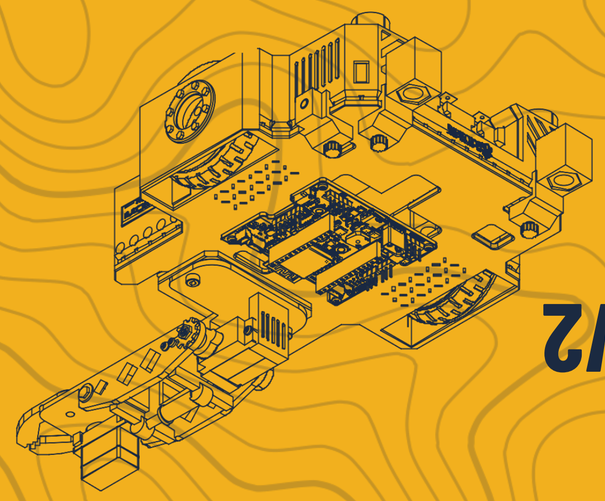


ROBOCORE V2

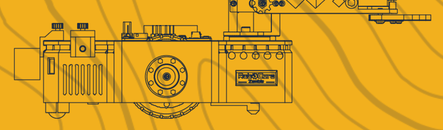
CLASSROOM ROBOT

Build. Code. Create.

- ESP32-S3 with Wi-Fi and Bluetooth
- Two encoder motors, four servo ports, IMU
- Line and ultrasonic sensing
- Arm module with gear-linked claw

DESIGNED IN ZAMBIA

ROBOCORE ZAMBIA · PLASTAL-BOT BUILDERS · LUSAKA



ROBOCORE V2

CLASSROOM ROBOT · REV 2.0 · DESIGNED IN ZAMBIA

RoboCore Zambia

PLASTAL-BOT BUILDERS

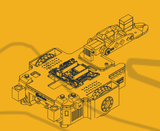
MEET ROBOCORE V2

A classroom robot designed and built in Zambia. The RoboCore V2 controller puts an ESP32-S3 with Wi-Fi and Bluetooth, two encoder motor drives, four servo ports, a motion sensor and sockets for line and ultrasonic sensing on one board, running from a single lithium cell with USB-C charging. The robot arrives assembled; the arm module bolts on in minutes.

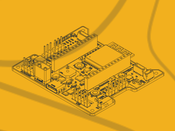
- **ESP32-S3 with Wi-Fi and Bluetooth**
- **Two encoder motors, four servo ports, IMU**
- **Line and ultrasonic sensing**
- **Arm module with gear-linked claw**



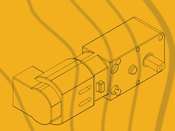
WHAT'S IN THE BOX



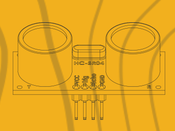
ROBOT, ASSEMBLED



ROBOCORE V2



2 ENCODER MOTORS



ULTRASONIC



LINE SENSOR



2 SERVOS + ARM

RoboCore V2 robot, assembled, with controller, HC-SR04 ultrasonic, 3-channel line sensor and battery holder · Arm module: base, boom, claw, 2 micro servos, 2 ballast clips · USB-C cable · Screw and nut pack · Quick-start card. 18650 cells and the M8 ballast nuts are not included.

SPECIFICATIONS

Controller	RoboCore V2, ESP32-S3, 68 x 60 mm
Power	1S Li-ion, 3 x 18650 in parallel, USB-C charging
Drive	2 x TT encoder motors, 56 mm wheels, 2 casters
Sensing	3-channel line sensor, HC-SR04 ultrasonic, IMU
Robot	167 x 166 x 65 mm, about 540 g with the arm
Box	273 x 203 x 83 mm, E-flute card

DOCS & CODE

780126 RC2.0

Not a toy. Contains small parts and screws: not for children under 8. Use only 1S lithium cells; charge through USB-C under supervision. Keep metal away from the antenna.

PLASTAL-BOT BUILDERS

DESIGNED IN ZAMBIA

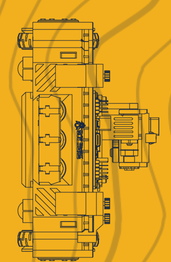
RoboCore Zambia · Lusaka · 2026 · Rev 2.0

ROBOCORE V2

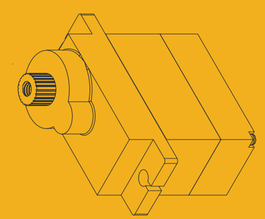
CLASSROOM ROBOT · REV 2.0

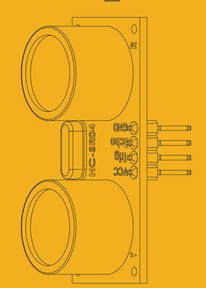
DESIGNED IN ZAMBIA

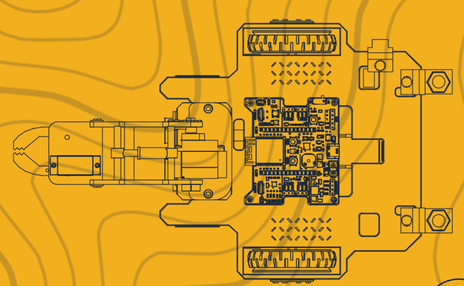
RC2.0 · 2026 · plastalbotbuilders.com











BUILD | CODE | CREATE

A robot a class can build, program and repair

RoboCore Zambia

OUTSIDE. RoboCore V2 mailer box, roll-end tuck-top, outer 273 x 203 x 83 mm, E-flute 1.5 mm. Print 1:1; navy solid = cut, grey dashed = fold. Bar = 100 mm.